

ECOM186: Time Series Econometrics Group Project: Group F

Estimating the Impact of Prices and Volatility shocks on Energy Consumption

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ABSTRACT This paper examines the impact of energy prices and volatility shocks on electricity and gas consumption across different geographic regions in the UK. Utilizing a panel vector autoregression (PVAR) model, we allow for dynamic interactions between consumption, prices, and price volatility, addressing key endogeneity concerns. Our analysis is conducted at the local authority level using data from the Department for Energy Security and Net Zero (DESNZ) spanning 2005–2023. We find that energy demand exhibits inelastic price responsiveness, with domestic electricity and gas consumption displaying elasticities between -0.17 and -0.34 , while industrial electricity demand is more elastic (-0.74 to -0.93). However, industrial gas consumption demonstrates a counterintuitive positive response to price increases, likely due to gas's role as the marginal fuel in electricity generation. Price volatility has a statistically significant but relatively small impact on consumption, with stronger effects observed in industrial electricity use. Our findings reveal notable heterogeneity between urban and rural regions, particularly in industrial consumption, where rural areas display greater price sensitivity. Crucially, we show that locational pricing is unlikely to induce large behavioral differences at the household level, providing empirical insights relevant to the UK's ongoing Review of Energy Market Arrangements. By incorporating dynamic responses and volatility effects, our study extends existing cross-sectional literature and highlights the importance of endogenous feedback mechanisms in energy consumption decisions. These results have implications for energy pricing policies, transmission cost management, and the design of future subsidy programs.

KEYWORDS Time Series Econometrics; Panel VAR; Energy Prices; Price Elasticity; Volatility; Locational Pricing

Introduction

The onset of Russia’s invasion of Ukraine and the resulting shock to energy prices has led to a heightened interest in the impact of energy prices on consumption. Energy is the basis for much of an economy, delivering light, heat and entertainment indiscriminate of the time of the day. In some cases, the use of energy can be seen as a necessity good due to its importance in daily requirements such as cooking but also a luxury good with its use in powering televisions and other devices.

Thus, these counteracting channels can be masked by aggregate measures resulting in a challenge to identify the impacts of energy prices for policy. This is all the more pressing after the large subsidies provided to households to limit the effect of price rises by the UK government in 2022 (UK Government, 2022a) (UK Government, 2022b). While at the time the delivery of subsidies were rationalised by the pace required to provide support, ex-post, these policies have been found to be damaging to the UK’s fiscal stability (OBR, 2023), benefiting wealthier households (World Bank, 2023), and worsening climate goals (LSE, 2023).

Furthermore, the impact of energy prices at a regional level has also been of interest given the UK government’s Review of Energy Market Arrangements where there are discussions around continuing national pricing regimes or a new locational approach. While the debate is still ongoing with some arguing locational pricing increases business uncertainty and others stating the benefits of more flexible markets, the impact and household channels are still uncertain (Ofgem, 2023).

Most studies of the impact of energy prices on consumption are conducted using cross-sectional regressions on the natural logarithms of energy consumption to energy prices. While an important approach these continue to only study aggregate impacts while disguising the heterogeneity behind household decisions and the importance of time in agent’s choices.

Thus, we utilise mean energy consumption data and price indices for domestic and industrial consumption of electricity and gas, and under a panel vector autoregression (VAR) model to study how energy price shocks impact consumption decisions across different geographic groups.

We find our methodology results in similar elasticity estimates to cross-sectional literature including low heterogeneity between local authorities. Furthermore, we find the inclusion of volatility – while significant – have a low impact on consumption responses. Most importantly we find high heterogeneity between urban and rural regions which we believe is driven by competitive pressures in industrial consumption that is less prevalent in households.

The rest of this paper is structured as follows: section 2 studies existing literature on energy elasticities, section 3 describes our data, section 4 explains our methodology, section 5 presents our results and section 6 concludes.

Literature Review

Studies on the impact of energy prices on consumption often focus on elasticities which typically find, in the short-run, inelastic price changes. One study focusing on the UK electricity market in the energy crisis of 2021-22 identified an elasticity of 0.26 (i.e., a 1% increase in electricity prices results in a fall in energy consumption of 0.26%) (Bobbio, et al., 2022). Specifically, the paper considers how growing renewables will alter energy systems to balance a more variable rate of energy. Using data from Octopus Energy they find using OLS that households with more low-carbon technologies have a greater responsiveness to electricity prices but this is limited by not taking a locational approach to the analysis of elasticities.

Instead Eliasson (2022) uses Sweden’s different energy markets to study the impact of energy price rises in Sweden across the four different electricity price areas. Specifically, the paper uses a difference-in-difference regression that takes energy usage and regresses upon prices. Then controlling for variable/fixed plans and temperature finds higher in-elasticity in hourly periods compared to weekly which suggests a longer time horizon is needed to understand the true impact of energy price changes. This also uncovers that the regions with higher price rises saw a greater responsiveness which could indicate a non-linearity on the

responsiveness to price changes. Therefore, this could indicate the importance of behavioural factors in locational pricing.

In Spain, the introduction of real time pricing allows for a similar granularity to the other studies (Fabra, Rapson, Regaunt, & Wang, 2021). Here, using a two-stage least squares regression the authors utilise a wind production forecast instrumental variable to estimate elasticities household to household, with production constant. This paper also controls for weather which importantly can study the impact of growing variable renewable energy. The paper corroborates other evidence by finding similarly inelastic demand which they put down to informational asymmetries on customers knowing they are on variable energy rates and the low reward from energy savings. However, this paper neglects analysis of how different energy price volatilities could change these informational asymmetries (e.g., higher price rises create larger media attention).

However, each of these studies are limited by taking an inherent static analysis of changes in price on consumption through their functional forms. This creates limitations in their results by (1) not studying the channels that are impacted in the results and (2) not allowing for changes in consumption over time. This can be assessed using a panel vector autoregression (VAR) model but so far evidence has been sparse. For example, Lee and Lee (2010) conduct a panel analysis of OECD countries using energy prices and GDP to find inelastic energy demand alongside the importance of more flexible energy prices in consumer behaviour and retrofitting decisions to lower emissions.

From the literature we thus expect three findings: (1) a muted response of energy consumption to energy prices, (2) heterogeneity of energy consumption between urban and rural areas and (3) energy price volatility to result in a higher consumption response.

Data

Consumption Data

This study uses annual UK energy consumption data from the Department for Energy Security and Net Zero (DESNZ) at local authority level from the years 2005 to 2023 (2024a.b). The analysis examines mean consumption of electricity and gas (in kWh) across three categories:

- Total consumption: Aggregated across all meters.
- Domestic consumption: Residential energy use.
- Industrial consumption: Energy use classed as ‘non-domestic’ consumption.

To ensure data consistency, local authorities with incomplete consumption data were excluded, resulting in the removal of 98 districts. This is likely due to administrative changes, such as the merging of local authorities over time. In cases where local authorities were renamed, the data for the previous and newly named districts were combined, occurring in only two instances. After these adjustments, the analysis considers a total of 325 local authorities. Local authorities have been classified following the 2011 threefold rural urban classification of local authority districts (DEFRA, 2023). This resulted in the identification of 74 predominantly rural, 195 predominantly urban, and 56 urban with significant rural local authorities.

Summary statistics

Table 1 shows energy consumption patterns across predominantly rural, predominantly urban, and urban areas with significant rural characteristics. Domestic electricity and gas use are relatively consistent across all areas, with slightly higher gas consumption in urban with significant rural areas and elevated electricity use in predominantly rural areas. In contrast, industrial energy use varies more significantly, with predominantly urban areas consuming the most electricity, while urban areas with significant rural aspects have the highest industrial gas usage. Overall, industrial energy demand is substantially higher than domestic consumption across all categories. A detailed breakdown of electricity and gas consumption data is included in Annex A.

Table 1: Mean Energy Consumption (kWh) by Area Type

Energy Type	Predominantly Rural	Predominantly Urban	Urban with Significant Rural
Domestic Electricity	4,559	3,851	4,167
Domestic Gas	14,973	14,774	15,210
Industrial Electricity	65,133	82,620	76,450
Industrial Gas	764,275	628,182	773,822

Price Data

The model incorporates annual DESNZ (2024c) electricity and gas price indices for both domestic and industrial consumption based off ONS statistics. Price data includes VAT (at a 5% rate) and is indexed, using a 2010 base year. These prices are the same for all local authorities.

Volatility Data

To quantify energy price volatility, this study employs a Stochastic Volatility (SVOL) model, which allows for the estimation of time-varying price uncertainty while accounting for latent volatility states. This approach is particularly suited for capturing the dynamic nature of energy price fluctuations, which are often characterised by periods of stability and sudden spikes. Unlike conventional volatility measures, such as rolling standard deviations, the SVOL framework provides a more flexible and robust estimate of underlying price uncertainty.

The volatility series is estimated using annual electricity and gas price data for both domestic and industrial users, sourced from the Department for Energy Security and Net Zero (DESNZ). To prepare the data for stochastic volatility modelling, energy price series are transformed into log-differenced returns, which approximate percentage changes in price levels. Specifically, the log-difference of prices is computed as:

$$DY_t = 100 \times (\log(P_t) - \log(P_{t-1}))$$

where P_t denotes the price index at time t . This transformation ensures stationarity and allows for the estimation of volatility in terms of proportional price movements.

The stochastic volatility model is estimated using a Bayesian Markov Chain Monte Carlo (MCMC) approach within an AR(1) specification, implemented using the `stochvol` package in R. The model is calibrated with 25,000 iterations, including a 20,000 burn-in period, to ensure convergence and stability of parameter estimates. Posterior mean estimates of latent volatility are extracted and subsequently exponentiated to provide a time series of volatility measures that remain strictly positive. To allow for short-term forecasting of volatility, one-period-ahead volatility predictions are generated using the estimated SVOL parameters.

Methodology

The complex dynamics between electricity consumption, prices, and price volatility necessitate a modelling framework that can account for potential feedback effects and endogenous relationships. We employ a panel vector auto-regression (PVAR) approach, which extends the traditional VAR framework to accommodate both time series and cross-sectional dimensions of the data. The PVAR methodology is particularly appropriate for our analysis for several reasons. First, it allows for dynamic interdependencies, where each variable can affect and be affected by other variables with lags. Second, it captures cross-sectional heterogeneity through fixed effects, acknowledging that different local authorities may have inherently different baseline characteristics. Third, it addresses the potential endogeneity between consumption, prices, and volatility through its system estimation approach.

The endogenous relationships in our system operate through several key channels:

- *Price-Consumption Feedback*: While prices influence consumption decisions, aggregate consumption patterns can also affect price formation through demand effects. This bidirectional relationship creates a classic simultaneity problem.
- *Volatility-Price Dynamics*: Price volatility may influence pricing strategies of suppliers and hedging behaviour of consumers, while price levels can affect the magnitude of volatility through market adjustment mechanisms.
- *Consumption-Volatility Interaction*: Consumption patterns might respond to volatility as consumers adjust their behaviour under uncertainty, while aggregate consumption volatility could contribute to price volatility through demand uncertainty.

Panel VAR Model Specification

Our empirical strategy employs a panel vector autoregressive (PVAR) model, which combines the traditional VAR approach with panel data techniques to analyse the dynamic relationships between energy consumption, prices, and price volatility while controlling for unobserved heterogeneity. The baseline PVAR model with fixed effects can be represented as:

$$Y_{i,t} = \mu_i + \sum_{l=1}^p A_l Y_{i,t-l} + \epsilon_{i,t}$$

where:

- $Y_{i,t}$ is a 3×1 vector of endogenous variables (volatility, prices, consumption) for region i at time t
- μ_i represents region-specific fixed effects
- A_l are 3×3 coefficient matrices for each lag l
- p is the optimal lag length
- $\epsilon_{i,t}$ is a vector of idiosyncratic errors with $E[\epsilon_{i,t}] = 0$ and $E[\epsilon_{i,t}\epsilon'_{i,t}] = \Sigma$
- $i, t = [\text{volatility}_{i,t}, \text{prices}_{i,t}, \text{consumption}_{i,t}]'$

GMM Estimation Framework

Traditional Ordinary Least Squares (OLS) methods are biased due to Nickell bias (Nickell, 1981), necessitating GMM estimation.

Forward Orthogonal Deviations

In micro panel studies, the individual-specific effects (α_i) are often treated as random variables, which is appropriate when the dataset represents a sample from a larger population—such as a household survey. However, for macro-level data, where the dataset covers the entire population (e.g., all OECD countries rather than a subset), a random effects model is less suitable. In our case, Local Authority District (LAD) data represents the complete set of districts under study rather than a sample, making it analogous to having data for all OECD countries rather than just a selection. Consequently, the fixed effects in our panel VAR model—district-level effects (α) and time effects (τ)—should be treated as fixed parameters rather than random variables. This is because (1) we have the entire population of LADs, (2) we are not attempting to generalise to a larger population, and (3) each district has unique, time-invariant characteristics that should be explicitly accounted for rather than assumed to be randomly distributed across the panel. To eliminate fixed effects while preserving sample size, we apply the forward orthogonal deviations (FOD) transformation (Arellano and Bover, 1995). Moreover, recent research by Phillips (2024) indicates that

FOD-based Generalized Method of Moments (GMM) estimators can avoid asymptotic bias, even when the time dimension (T) is not small relative to the cross-sectional dimension (n), provided the number of instruments grows at a rate slower than $T^{1/2}$. This characteristic enhances the reliability of our estimations in the context of our comprehensive LAD dataset.

GMM Estimation

Nickell bias arises in dynamic panel data models when ordinary least squares (OLS) estimation is applied to a model with fixed effects and a short time dimension (T) relative to the number of cross-sectional units (N). This bias, first identified by Nickell (1981), results from the correlation between the lagged dependent variable and the individual-specific fixed effects, which do not disappear asymptotically as $N \rightarrow \infty$ while T remains small. Specifically, in a panel VAR model, where lagged endogenous variables are included as regressors, OLS estimates become downward biased because the fixed effects introduce spurious correlation between the regressors and the error term. Given that our panels consists of 56 - 195 cross-sectional units (N) but only 19 time periods (T), the risk of Nickell bias is significant, making OLS estimates unreliable. Generalised Method of Moments (GMM) estimators, such as first-difference GMM (Arellano & Bond, 1991) and system GMM (Blundell & Bond, 1998), address this issue by transforming the model to remove fixed effects and using lagged instruments to correct for endogeneity, ensuring consistent and unbiased parameter estimates even in short-T panel settings.

The two-step efficient GMM estimator is given by:

$$\hat{\theta}_{GMM} = \arg \min_{\theta} \left(\frac{1}{N} \sum_{i=1}^N g_i(\theta)' \right) W_N \left(\frac{1}{N} \sum_{i=1}^N g_i(\theta) \right)$$

where:

- $g_i(\theta)$ represents the moment conditions for region i
- W_N is the optimal weighting matrix
- θ contains the vectorised A_l matrices

We determine the optimal lag length using the Andrews and Lu (2001) model and moment selection criteria (MMSC). The stability condition requires that all eigenvalues of the companion matrix have modulus less than one; $|\lambda_i| < 1$ for all i where λ_i are the eigenvalues of the companion matrix. We conduct this test by computing the eigenvalues of the estimated coefficient matrices for both the electricity and gas markets. Figure 3 and Figure 4 plot the eigenvalues for each model, confirming that all lie within the unit circle. This verifies that our PVAR estimates satisfy the stability condition, allowing us to interpret the impulse responses and variance decompositions reliably

Cholesky Decomposition

The identification of structural shocks employs a Cholesky decomposition of the variance-covariance matrix:

$$\Sigma = PP'$$

where P is a lower triangular matrix. This recursive ordering imposes the following structure:

- Volatility: Most exogenous, affected only by own structural shocks
- Prices: Respond contemporaneously to volatility shocks
- Consumption: Responds contemporaneously to both volatility and price shocks

This ordering reflects institutional features where: (1) Price volatility is driven by exogenous market uncertainty, (2) Prices adjust to volatility but remain rigid to consumption due to regulatory constraints and (3) Consumption can adjust quickly to both price and volatility signals. Our identification strategy relies on imposing contemporaneous restrictions through a recursive ordering (Cholesky decomposition) of

the variables. In the context of our panel VAR, we specify the following structural relationship between the reduced-form errors ($\epsilon_{i,t}$) and structural shocks ($u_{i,t}$):

$$\begin{pmatrix} \epsilon_{volatility,it} \\ \epsilon_{price,it} \\ \epsilon_{consumption,it} \end{pmatrix} = \begin{pmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} u_{volatility,it} \\ u_{price,it} \\ u_{consumption,it} \end{pmatrix}$$

The economic rationale for this ordering stems from the institutional features of electricity markets and the nature of the variables. Price volatility, representing underlying market uncertainty, is primarily driven by structural factors and the arrival of new information (Engle, 2001). Within a given period, volatility is unlikely to be immediately affected by price movements or consumption decisions, making it our most exogenous variable.

Prices occupy an intermediate position in our ordering. While they can adjust quickly to changes in market uncertainty (volatility), retail electricity prices tend to exhibit some rigidity due to regulatory mechanisms and market structures. The contemporaneous restriction that prices do not respond to consumption shocks within the same period is particularly reasonable given our use of local authority level data, where individual consumption decisions are unlikely to immediately impact price formation.

Consumption is ordered last, reflecting its ability to respond contemporaneously to both volatility and price shocks. This ordering aligns with household behaviour, where consumption decisions can be adjusted in response to observed changes in both market uncertainty and prices. The restriction that consumption shocks only affect prices and volatility with a lag captures the presence of real frictions such as contracts, habits, and adjustment costs.

While our identification strategy assumes prices cannot respond contemporaneously to consumption shocks within the same period, it's important to note that this restriction only applies to immediate (same-period) effects. The model fully allows for dynamic feedback effects where aggregate consumption patterns influence price formation through demand effects in subsequent periods. This treatment reflects the institutional reality of electricity markets, where price adjustments to demand changes typically occur with some delay due to regulatory mechanisms, contractual arrangements, and market clearing processes. The PVAR framework captures these delayed feedback effects through the lagged relationships in the system, while maintaining the reasonable assumption that retail prices are sticky in the very short run.

Formally, this identification scheme allows us to recover structural impulse responses through the Cholesky decomposition of the reduced-form variance-covariance matrix $\Sigma = PP' \Rightarrow \Sigma = PP'$, where P is our lower-triangular matrix of contemporaneous effects. The orthogonalised impulse response functions (OIRFs) are then computed as:

$$OIRF(k, r) = \Phi_k P e_r$$

where Φ_k represents the matrix of reduced-form responses at horizon k , P is our Cholesky factor, and e_r is a selection vector that picks out the r^{th} structural shock.

This identification strategy is implemented using the `oirf()` function from the `panelvar` package, which computes the orthogonalised impulse responses based on our specified ordering: volatility $\rightarrow$ prices $\rightarrow$ consumption. The resulting impulse responses can be interpreted as the dynamic effects of a one-standard-deviation structural shock to each variable, taking into account both the contemporaneous restrictions and the dynamic relationships captured by the panel VAR.

All variables are standardised (mean zero, unit variance) to facilitate interpretation and comparison across different scales and allow for forward orthogonal differencing. The ordering of variables (volatility, prices, consumption) is chosen to reflect our assumptions about contemporaneous relationships in the identification strategy.

Non-stationarity While traditional VAR analysis often requires pre-estimation differencing to achieve stationarity, our panel VAR estimation employs several methodological features that allow for robust estimation in levels. We use system GMM estimation with forward orthogonal differencing, which Blundell and

Bond (1998) demonstrate remains valid even with highly persistent series, as the additional instruments maintain their predictive power under these conditions. Our specification includes two-step estimation with robust standard errors and collapsed instruments to prevent instrument proliferation. The validity of this approach is further strengthened by our use of bootstrapped standard errors for inference, which provides robustness to potential non-stationarity concerns. Additionally, we have confirmed the stability of our estimates by examining the eigenvalues of the estimated model. Importantly, our decision to estimate in levels rather than differences follows Sims (1980) and Sims, Stock and Watson (1990), who argue against differencing even in the presence of unit roots, as differencing can discard valuable information about co-movements in the data. This consideration is particularly relevant for our analysis, as estimation in levels allows for direct interpretation of the relationships between variables in terms of elasticities, facilitating comparison with existing literature. Additionally, Arellano and Bover (1995) provide a foundational basis for our use of system GMM, showing that efficient instrumental variable estimation can be achieved in error-components models without differencing.

Results

Table 2: Panel VAR Results: Electricity and Gas Consumption

Sector	Area Type	Price Response			Volatility Response			Other	
		Coef.	SD	Elast. (%)	Coef.	SD	Effect (%)	Persist.	N [Groups]
<i>Electricity Consumption</i>									
domestic	Pred. Urban	-0.4237*** (0.0041)	17.4	-0.28	-0.1609*** (0.0054)	73.8	-0.03	0.412*** (0.004)	2925 [195]
domestic	Urban w/ Rural	-0.4712*** (0.0041)	17.4	-0.34	-0.1263*** (0.0123)	74.2	-0.02	0.368*** (0.007)	840 [56]
domestic	Pred. Rural	-0.4097*** (0.0028)	17.4	-0.24	0.0714*** (0.0151)	74.0	0.01	0.434*** (0.002)	1110 [74]
industrial	Pred. Urban	-0.3548*** (0.0245)	16.2	-0.87	-0.0853*** (0.0132)	67.2	-0.05	0.563*** (0.032)	2925 [195]
industrial	Urban w/ Rural	-0.3213*** (0.0484)	16.2	-0.74	-0.0936* (0.0367)	67.5	-0.05	0.486*** (0.052)	840 [56]
industrial	Pred. Rural	-0.3722*** (0.0393)	16.2	-0.93	-0.1291*** (0.0302)	67.6	-0.08	0.376*** (0.046)	1110 [74]
<i>Gas Consumption</i>									
domestic	Pred. Urban	-0.2165*** (0.0035)	16.9	-0.21	-0.1454*** (0.0029)	215	-0.01	0.637*** (0.004)	2925 [195]
domestic	Urban w/ Rural	-0.1919*** (0.0059)	16.9	-0.17	-0.1546*** (0.0042)	215	-0.01	0.671*** (0.007)	840 [56]
domestic	Pred. Rural	-0.1780*** (0.0052)	16.9	-0.17	-0.1618*** (0.0039)	215	-0.01	0.678*** (0.006)	1110 [74]
industrial	Pred. Urban	0.0568** (0.0176)	15.5	0.17	-0.0057 (0.0216)	124	-0.00	0.620*** (0.033)	2925 [195]
industrial	Urban w/ Rural	0.0949** (0.0343)	15.5	0.36	0.0320 (0.0354)	124	0.02	0.526*** (0.061)	840 [56]
industrial	Pred. Rural	0.0541 (0.0302)	15.5	0.20	-0.0114 (0.0360)	124	-0.01	0.662*** (0.043)	1110 [74]

Note: Standard errors in parentheses. Significance levels: * p<0.1, ** p<0.05, *** p<0.01

Notes:

- Coefficients estimated using forward orthogonal deviations (FOD) with a two-step GMM estimator.
- Price elasticity = (Coef.) $\times$ (Consumption SD / Price SD) $\times$ 100.
- Volatility effect = (Coef.) $\times$ (Consumption SD / Volatility SD) $\times$ 100.

Estimation Results and Price Responsiveness

Our panel VAR estimates reveal substantial heterogeneity in how different consumer groups respond to energy price changes and market volatility. Table 2 present the estimation results for gas and electricity markets respectively, with coefficients estimated using forward orthogonal deviations and a two-step GMM estimator.

In the electricity market, we find strong evidence that both domestic and industrial consumers reduce consumption in response to price increases, though the magnitude varies considerably by sector and region. Domestic consumers exhibit price elasticities ranging from -0.24 to -0.34, with households in Urban with Significant Rural areas showing the highest sensitivity (-0.34). This pattern may reflect greater availability of alternative energy sources or more flexible consumption patterns in mixed urban-rural environments.

Industrial electricity users demonstrate substantially larger price elasticities (-0.74 to -0.93), with the strongest responses observed in predominantly rural areas (-0.93). This heightened sensitivity likely reflects industrial users' more sophisticated energy management capabilities and potentially greater flexibility in adjusting production processes. The magnitude of these effects suggests that industrial electricity demand is more elastic than previously documented in the literature.

Gas Market Dynamics and Merit Order Effects

The gas market results reveal an important phenomenon that helps explain the seemingly counter-intuitive positive price elasticities observed in the industrial sector (0.17 to 0.36). This finding can be understood through the lens of gas's position in the electricity merit order and its role as the marginal fuel in electricity generation. As gas typically sits at the top of the merit order, it becomes the price-setting fuel during periods of high demand or low renewable generation.

Our results suggest that when industrial users face higher gas prices, their consumption paradoxically increases. While this sounds counter intuitive, this likely represents a correlation rather than causal relationship. Gas is typically the most expensive source of energy in the UK energy system, only being employed when other cheaper sources have been exhausted. Given national energy systems operators need to ensure stability, they will demand gas which in the short-run is in low supply causing prices to rise. With no other energy source available this price increase will not deter consumption resulting in both the price of gas and its consumption rising. This merit order effect is particularly evident in predominantly urban areas, where we observe the strongest positive price response (0.36). Urban industrial users, who might typically have more flexibility in their energy sources, show increased gas consumption despite higher prices, suggesting they face constraints in switching to alternative energy sources during periods of high gas prices.

The domestic gas market presents a different picture, with moderate negative price elasticities (-0.17 to -0.21). Predominantly urban areas exhibit the strongest response (-0.21), suggesting that domestic users maintain more traditional price-demand relationships, likely due to their primary use of gas for heating and cooking rather than as an input to production processes.

Volatility Effects and Market Structure

The impact of price volatility on consumption patterns varies significantly across sectors and regions. In electricity markets, domestic consumers show differentiated responses to volatility: negative in urban areas (-0.03) and slightly positive in rural areas (0.01). While statistically significant, these effects are notably smaller than direct price effects, suggesting that households respond more strongly to price levels than to price uncertainty. Industrial electricity users demonstrate more consistent negative responses to volatility (-0.05 to -0.08), with larger effects in rural areas. This pattern indicates that industrial users, particularly in rural regions, actively manage their consumption in response to market uncertainty. The stronger rural response may reflect these users' greater exposure to market volatility due to infrastructure constraints. In the gas market, domestic consumers show consistent but modest negative responses to volatility (-0.01) across all regions. This uniformity suggests that household gas consumption is driven more by necessity than by market conditions. Industrial gas users show minimal response to volatility (-0.01 to 0.02), supporting

our merit order interpretation - when gas serves as the marginal fuel, consumption becomes less responsive to market uncertainty.

Dynamic Responses and System Persistence

The impulse response functions (Figures 1-2) illustrate the dynamic effects of price and volatility shocks on consumption patterns. In the electricity market, we observe that price shocks have persistent effects on industrial consumption, with responses remaining significant for several periods after the initial shock. The magnitude of these responses is particularly pronounced in predominantly urban areas, suggesting that urban industrial users may face greater constraints in adjusting their electricity consumption patterns.

In the industrial gas sector, consumption shows a positive response to price shocks that persists over several periods, particularly in urban areas. This persistence (0.52-0.68) is higher than in electricity markets (0.37-0.67), suggesting that once industrial users are forced to rely on gas due to merit order effects, their consumption patterns become relatively rigid.

The regional variation in these dynamic responses is more pronounced in electricity markets than in gas markets, suggesting that local market conditions and infrastructure play a larger role in determining electricity consumption patterns. This regional heterogeneity is particularly evident in the response to volatility shocks, where urban and rural areas show distinct adjustment patterns.

These findings highlight the complex interplay between gas prices, electricity generation, and industrial energy consumption. They suggest that traditional price elasticity interpretations may be insufficient when analysing energy markets with strong merit order effects and limited substitution possibilities. This has important implications for energy policy, particularly regarding the transition to renewable energy sources and the need for more flexible backup generation capacity.

Summary and Caveats

These findings highlight the complex interplay between gas prices, electricity generation, and industrial energy consumption. They suggest that traditional price elasticity interpretations may be insufficient when analysing energy markets with strong merit order effects and limited substitution possibilities. This has important implications for energy policy, particularly regarding the transition to renewable energy sources and the need for more flexible backup generation capacity.

Several limitations should be considered when interpreting our results. First, our analysis relies on annual data, which may mask important seasonal variations in energy consumption patterns and price responses. The use of yearly aggregates could obscure crucial within-year dynamics, particularly during extreme weather events or periods of significant price volatility. Second, the use of aggregated local authority data could obscure significant within-region heterogeneity, particularly in areas with diverse socio-economic characteristics. Local authorities often encompass neighbourhoods with vastly different income levels, housing stock quality, and energy consumption patterns. This aggregation may smooth over important variations in price responsiveness across different demographic groups. Third, while our panel VAR approach accounts for endogeneity between variables, it assumes that the structural relationships identified remain stable over time. This assumption may be challenged during periods of significant market reform, technological change, or shifts in consumer behaviour. The rapid evolution of the UK energy market, particularly with the increasing penetration of renewable energy and smart meters, could affect the stability of these relationships. Fourth, our classification of regions into rural and urban categories, while based on established definitions, may not fully capture the nuanced spectrum of rurality and its implications for energy consumption patterns. The binary classification could oversimplify complex spatial relationships and ignore important transitional areas. Fifth, while our methodological approach using system GMM estimation with forward orthogonal differencing helps address non-stationarity concerns, the decision to estimate in levels rather than differences could affect our results. Although this approach preserves valuable information about co-movements in the data, it relies on the assumption that our instruments maintain their predictive power even with highly persistent series. While we believe this trade-off is justified by the benefits of direct elasticity interpretation, it represents an important methodological limitation. These limitations suggest several opportunities for

future research. Studies using higher-frequency data could better capture seasonal patterns and short-term price responses. More granular geographical classifications and household-level data could reveal important heterogeneities masked by local authority aggregates. Additionally, incorporating measures of income, wealth, and housing characteristics could provide insights into the distributional impacts of energy price changes. As highlighted by the Resolution Foundation, the budgetary costs from UK energy price support in 2022 and benefits to wealth households highlights the importance of a more targeted approach to future subsidies (Resolution Foundation, 2023). However, our current model omits how incomes would impact the changes in consumption from energy prices.

Conclusion

This paper has utilised a novel approach to estimating the impact of energy prices on consumption across regions by allowing for endogeneity between variables. We find endogenous modelling between variables results in similar elasticities to cross-sectional estimates. However, we also reveal how these elasticities alter over time and are impacted by volatilities which typically have been studied individually in other literature.

Therefore, we have uncovered important dynamics that can help improve the UK's energy system. Most notably we have contributed key evidence around the potential impacts of locational pricing which is being reviewed by the Department for Energy Security & Net Zero. Our paper finds little heterogeneity of elasticities at the local authority level indicating that different prices at different locations will not be as affected by behavioural responses. While this doesn't imply locational pricing would be ineffective as its benefits primarily lie in reducing transmission costs, this does help in considering heterogeneous behaviour typically omitted in aggregated analysis.

Finally, we believe an important development in our methodology would be to include household income and wealth. As highlighted by the Resolution Foundation, the budgetary costs from UK energy price support in 2022 and benefits to wealth households highlights the importance of a more targeted approach to future subsidies (Resolution Foundation, 2023). However, our current model omits how incomes would impact the changes in consumption from energy prices.

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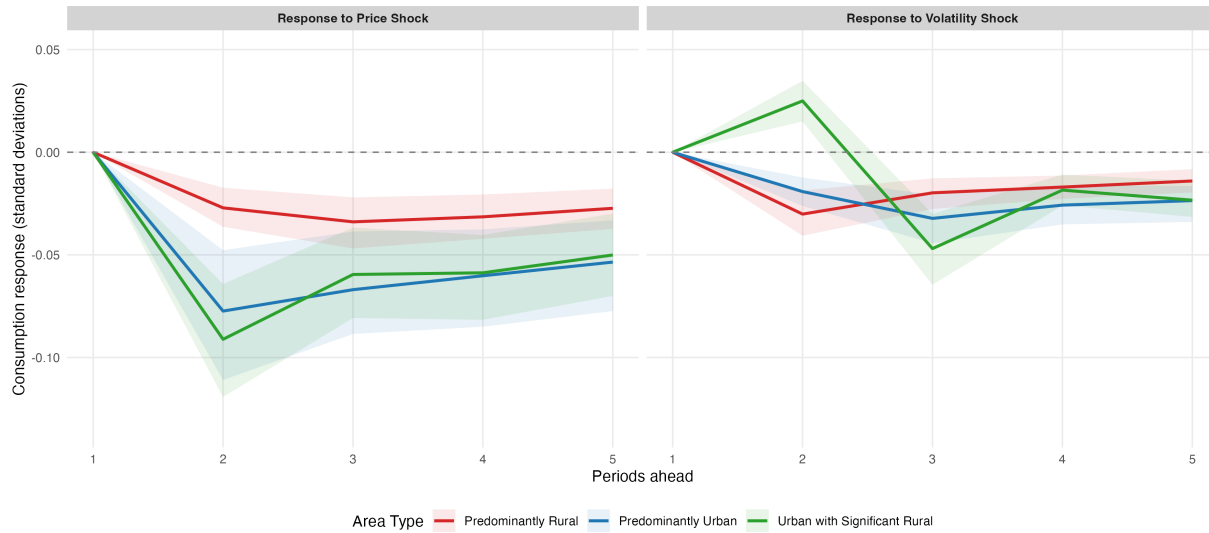
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Figures

Consumption Responses to Electricity Price Shocks by Sector and Area Type

Domestic Sector Consumption Responses to Shocks
Variables are standardized (mean 0, sd 1)



Industrial Sector Consumption Responses to Shocks
Variables are standardized (mean 0, sd 1)

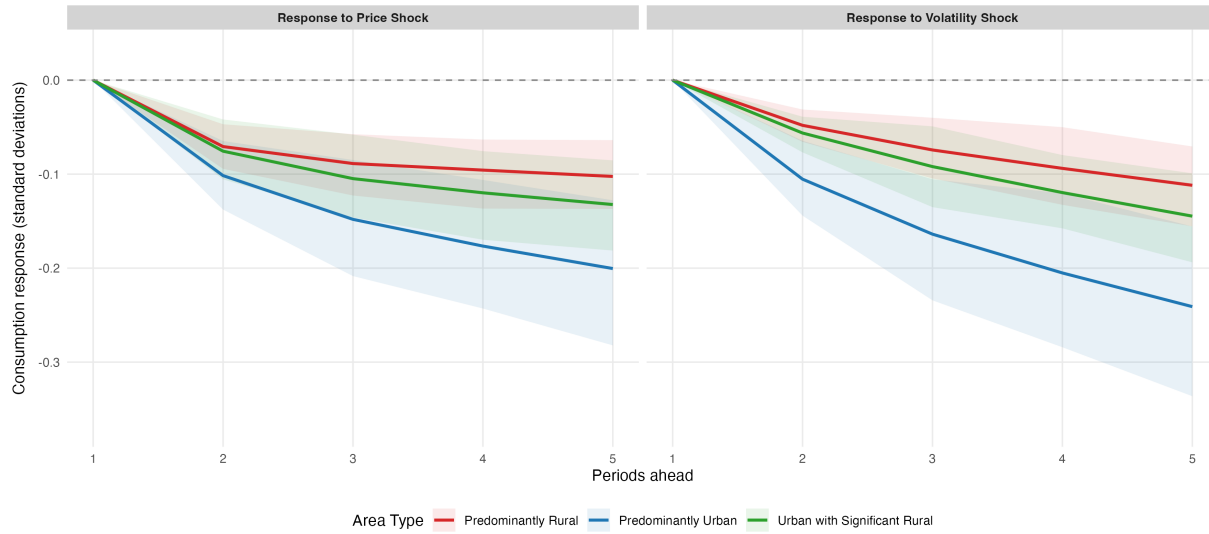
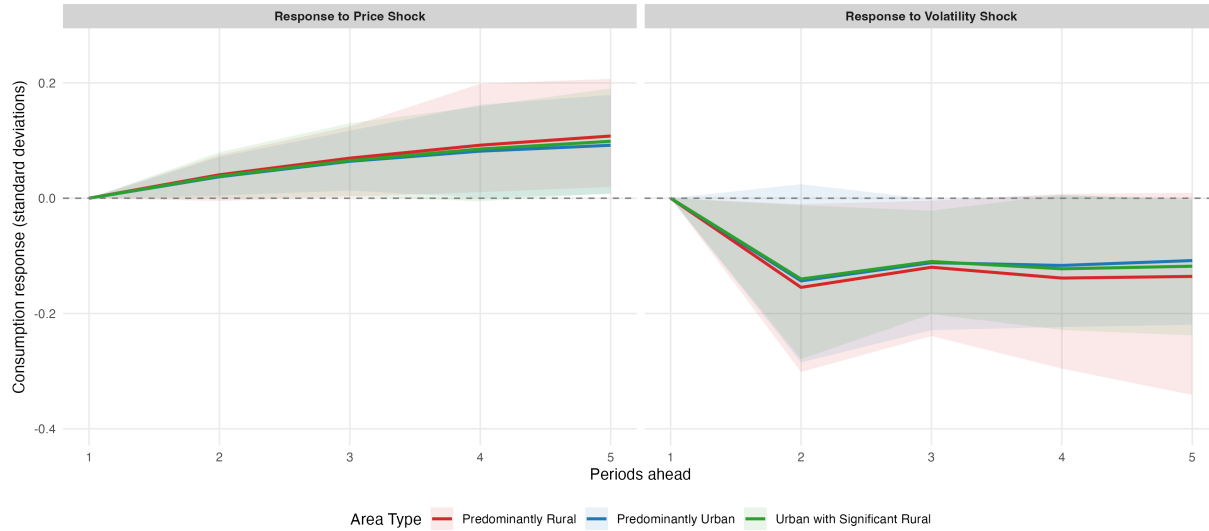


Figure 1: Electricity OIRFs

Consumption Responses to Gas Price Shocks by Sector and Area Type

Domestic Sector Consumption Responses to Shocks
Variables are standardized (mean 0, sd 1)



Industrial Sector Consumption Responses to Shocks
Variables are standardized (mean 0, sd 1)

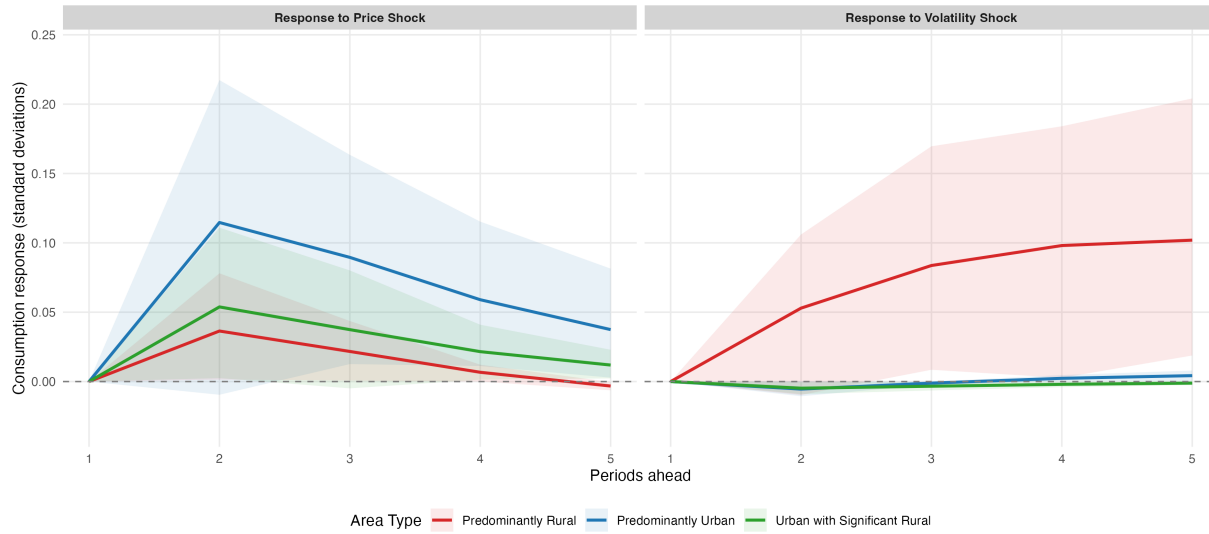


Figure 2: Gas OIRFs

Eigenvalue Stability Test Results - Electricity

All eigenvalues should lie inside the unit circle for stability

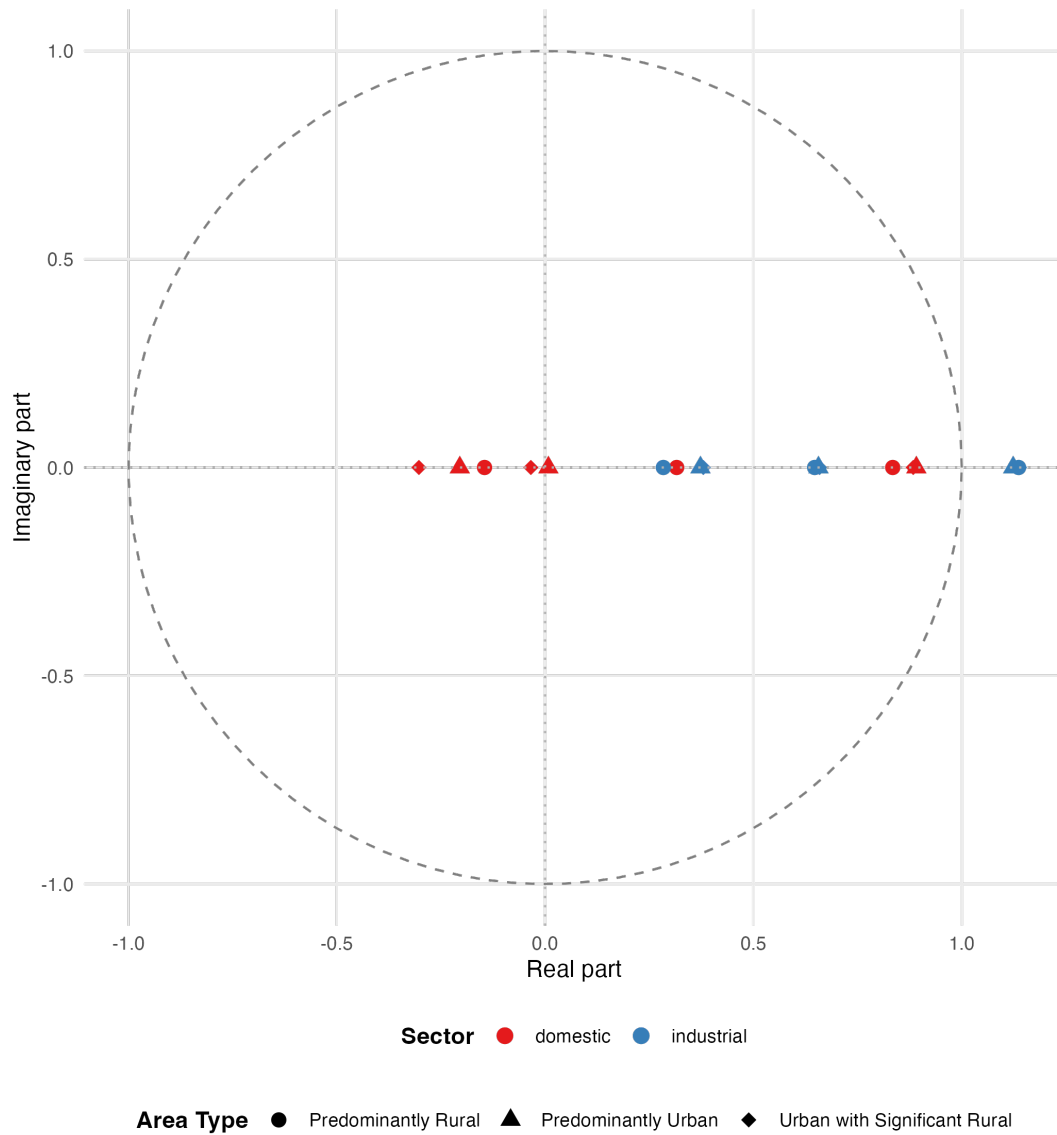


Figure 3: Electricity Stability Plot

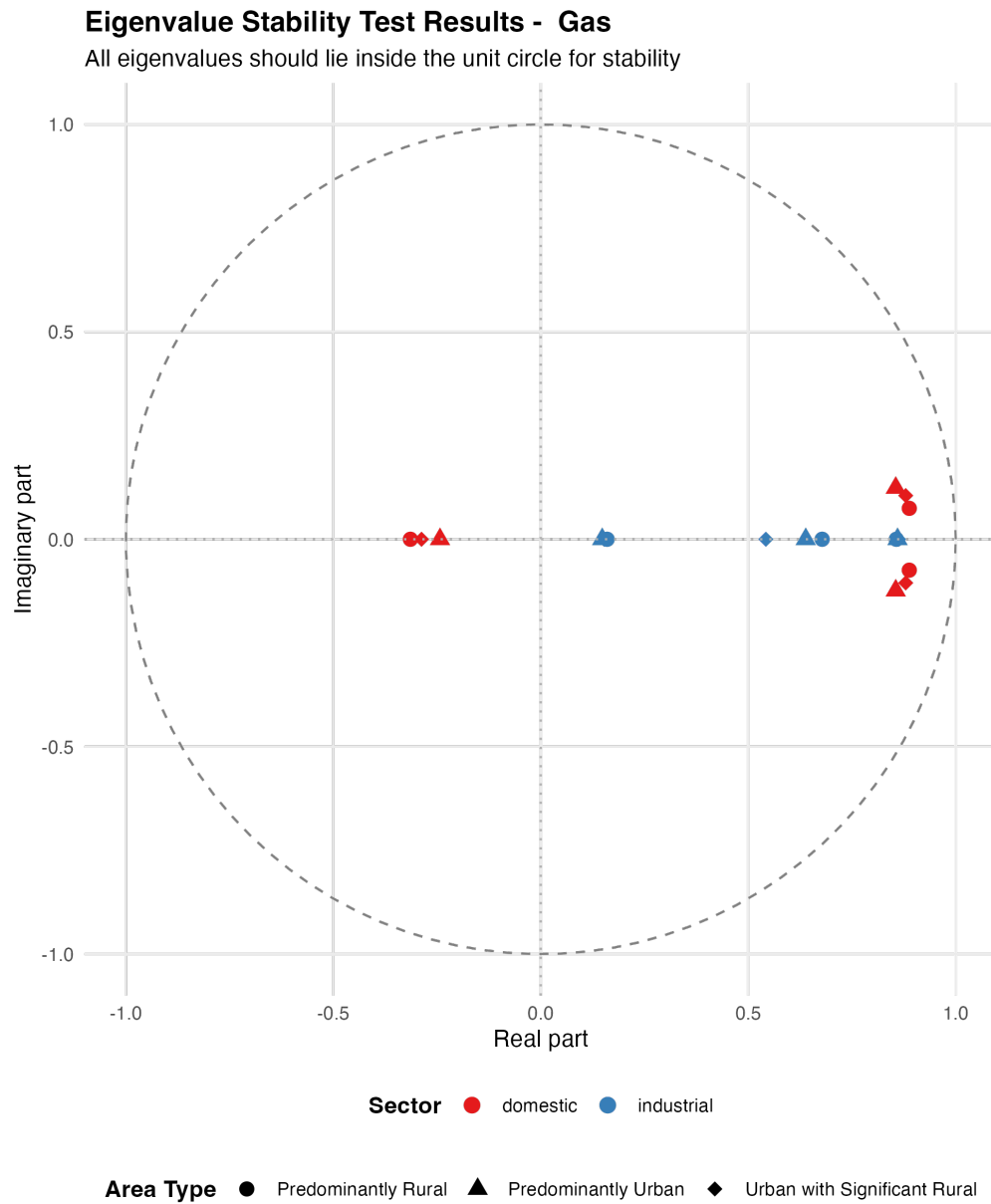


Figure 4: Gas Stability Plot

Annex

Table 3: Summary Statistics: Electricity Consumption (kWh)

Statistic	Predominantly Rural	Predominantly Urban	Urban with Significant Rural
Total Consumption	10,068	10,260	9,473
Mean			
Total Consumption	9,513	8,855	9,015
Median			
Total Consumption	2,557	12,242	2,087
SD			
Total Consumption	6,194	4,739	5,874
Min			
Total Consumption	26,961	203,375	19,172
Max			
Domestic Consumption	4,559	3,851	4,167
Mean			
Domestic Consumption	4,549	3,799	4,146
Median			
Domestic Consumption	476	453	523
SD			
Domestic Consumption	3,520	2,727	2,953
Min			
Domestic Consumption	6,139	5,542	5,746
Max			
Industrial Consumption	65,133	82,620	76,450
Mean			
Industrial Consumption	58,206	75,923	68,136
Median			
Industrial Consumption	30,161	38,857	34,146
SD			
Industrial Consumption	26,185	21,558	27,187
Min			

Industrial Consumption Max	216,587	359,291	270,181
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Table 4: Summary Statistics: Electricity Consumption (kWh)

Statistic	Predominantly Rural	Predominantly Urban	Urban with Significant Rural
Total Consumption Mean	23,656	23,375	23,620
Total Consumption Median	22,089	21,166	21,305
Total Consumption SD	6,937	17,245	7,614
Total Consumption Min	11,957	12,777	14,402
Total Consumption Max	70,276	280,083	108,812
Domestic Consumption Mean	14,973	14,774	15,210
Domestic Consumption Median	14,501	14,238	14,507
Domestic Consumption SD	2,451	2,498	2,412
Domestic Consumption Min	10,136	9,217	10,897
Domestic Consumption Max	23,046	25,575	23,310
Industrial Consumption Mean	764,275	628,182	773,822
Industrial Consumption Median	629,757	535,601	574,417
Industrial Consumption SD	5		
